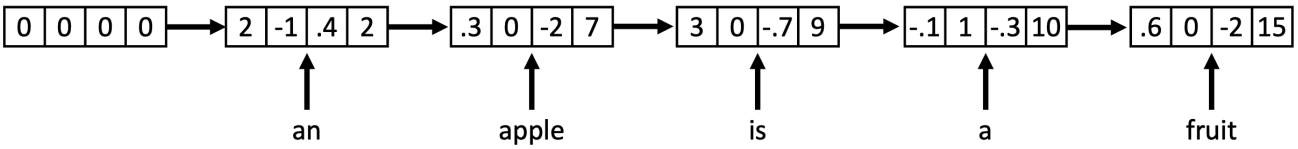
(N) A Trip Down Memory Lane (1/4) [10 Points]

When you read a sentence, there's a lot of information to keep track of. Consider this sentence:

The artist who lives in the tower painted a portrait of himself.

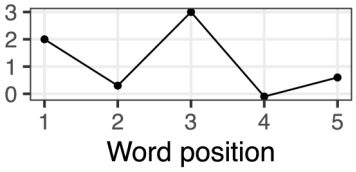
To understand who ***himself*** refers to, you need to remember the word ***artist*** from early in the sentence.

If you were building a machine that processed language, how would you get it to remember such information? One strategy is to have the system store information in a vector (a list of numbers). In the example below, the system starts out with a vector containing only zeroes (on the far left). It then reads in a sentence, one word at a time. After seeing each word, it updates the vector using some mathematical operations (for this problem, it's not important to know what those operations are).



For instance, after seeing ***an***, the system updates the vector of zeroes so that it becomes [2, -1, .4, 2]. Then, after seeing ***apple***, it updates [2, -1, .4, 2] so that it now reads [.3, 0, -2, 7]. In this way, the vector acts as a sort of memory that encodes information about everything the system has seen so far.

This type of system is called a **recurrent neural network**, and each position in its vector representation is called a **unit**. Since the system shown above uses a vector of size 4, it has 4 units. Each unit has some value at each point in the sentence. For instance, the value of the first unit goes from 0 to 2 to .3 to 3 to –.1 to .6. If we were to plot this unit's values over time, it might look something like the image to the right (we have left out the value of 0 at the start, since no information is available yet at that point).



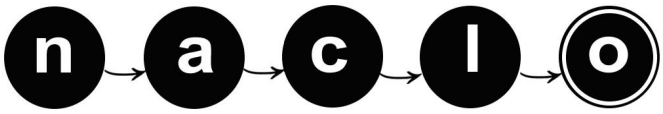
What do the numerical values inside a unit mean? It turns out that, most of the time, they don't have a clear meaning! Recurrent neural networks are designed to learn from data, so whatever they represent in their units is something that they themselves have learned, not something that a human designed. This means that there is no guarantee that a given unit's values will make sense to humans.

Occasionally, however, there is a unit that can be interpreted in a systematic way. In the example above, the second unit has a value of –1 after seeing ***an***, a value of 1 after seeing ***a***, and 0 otherwise. Thus, it seems like this unit has the special role of indicating whether the word that was just seen is ***a*** or ***an***.

Why might the system encode this information? This particular system was designed for the goal of predicting what word will come next in the sentence. For this goal, it is useful to know whether ***a*** or ***an*** has just appeared: If we have just seen ***a***, then the next word must be something that starts with a consonant, and if it is ***an***, then the next word must start with a vowel.

N1. What information is represented by system's fourth unit (the one with the values 0, 2, 7, 9, 10, 15)? Write your answer in your Answer Sheet.

<u>Make sure you record your answers in your Answer Sheets!</u>

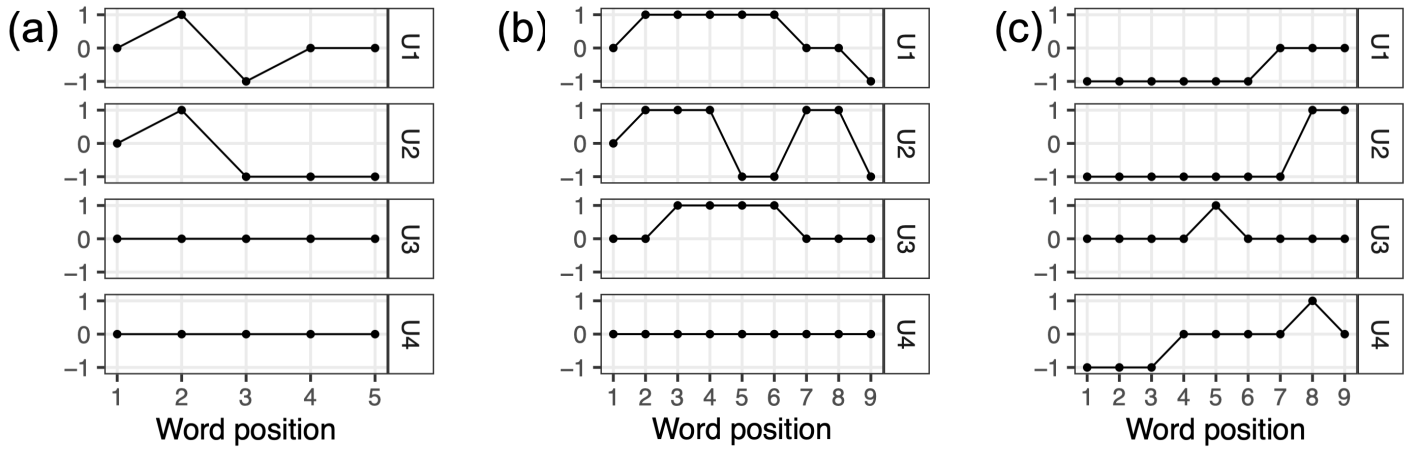


(N) A Trip Down Memory Lane (2/4)

N2. Researchers at NACLO Labs are analyzing a recurrent neural network that uses many units to store information about a sentence. They have identified four units—labeled U1, U2, U3, and U4—that seem to store interesting information. Below are nine sentences, labeled 1 through 9. On the next page are nine plots showing the values of U1 through U4 for these nine sentences—but the plots are in random order! On your Answer Sheets, match each sentence to the plot that goes with it.

1. The dog and the ravens support the happy yaks
2. The American knows the very energetic dog is smart
3. The raven that the crows admire visits the poet
4. This surprised the tall ostriches that supported the horse
5. The dogs near the owl that sings are Canadian
6. The happy penguins support the zebras that invented this
7. She supports the horse that the fast zebra entertains
8. The speedy sloth knows the dog and the yak
9. The impressive yaks meet the dog and the gazelles

N3. Here are three more plots. For each one, write a sentence that could have generated that plot (there are many correct answers; you only need to provide one for each plot).



N4. For each of the following sentences, complete the plots on your Answer Sheets to indicate the values that U3 would have:

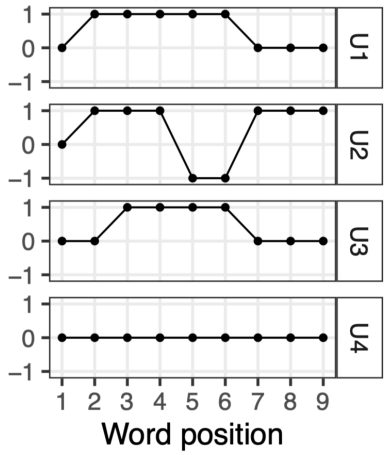
(a) The story that she told my cousin was exciting
(b) The story that she told to my cousin was exciting

<u>Make sure you record your answers in your Answer Sheets!</u>

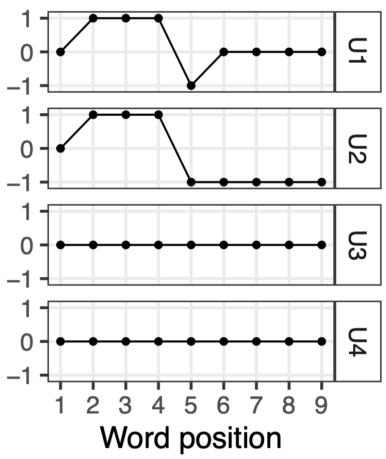
(N) A Trip Down Memory Lane (3/4)

These images are the choices for question N2 on the previous page.

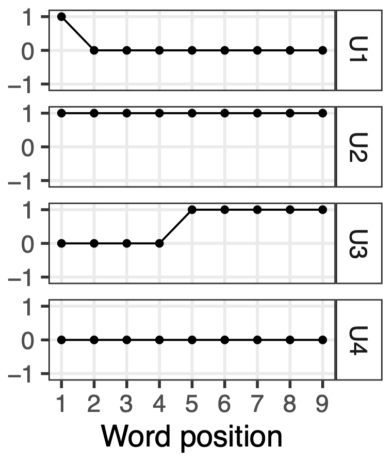
Option A



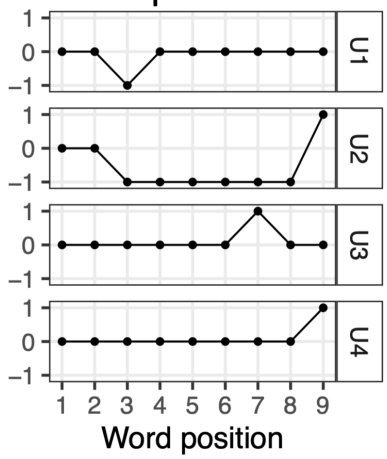
Option B



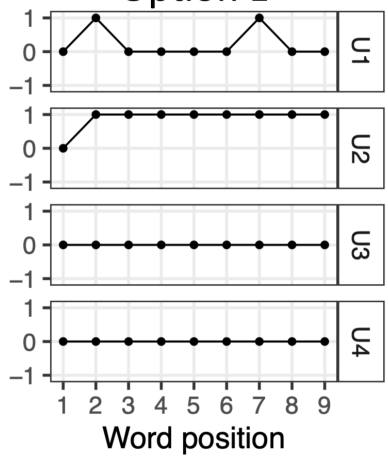
Option C



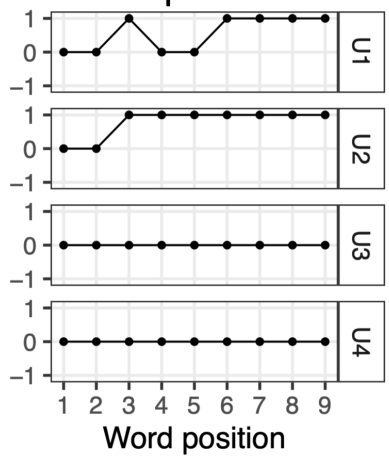
Option D



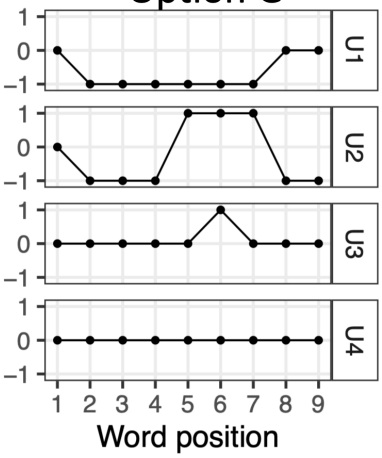
Option E



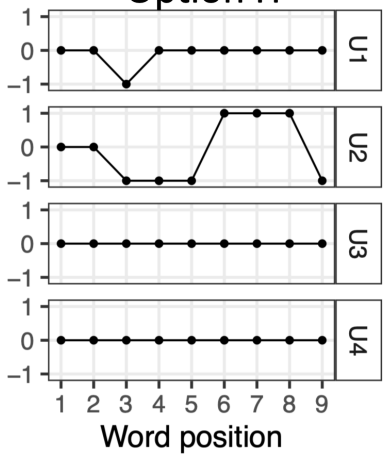
Option F



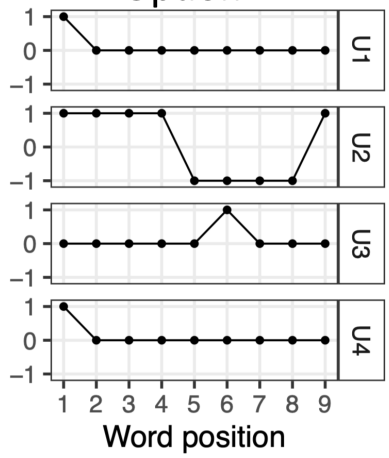
Option G

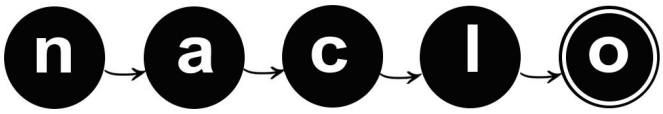


Option H



Option I





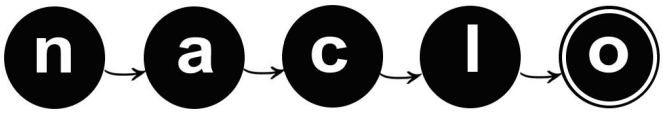
(N) A Trip Down Memory Lane (4/4)

Remember that this system's goal is to predict the next word based on the words it has seen so far; this is what it uses its vectors for. You decide to take a closer look at how well the system does at predicting the next word. To do this, you give the model the start of a sentence and analyze what words it thinks are likely to come next. For this experiment, you restrict the model's options so that the only words it can predict are ***is*** and ***are***. With this restriction, if the sentence starts with ***the dog*** (for instance), then you can be 100% sure that the next word must be ***is*** (creating ***the dog is***) rather than ***are*** (which would produce ***the dog are***).

What you are especially interested in is how individual units contribute to the system's predictions. To understand this, you focus on U1 and U2 and see what happens when you "turn off" each unit so that the model no longer has access to whatever that unit encodes. The results of these tests are in the table below. For example, let's look at the first row, which deals with the case where the sentence starts with ***the dogs***. In this case, the next word must be ***are***, not ***is***. When both U1 and U2 are kept on, the model gets this prediction right. It continues to get the answer right when U1 is off and U2 is on, or when U1 is on and U2 is off. However, when both U1 and U2 are turned off, the model's performance drops, as shown in the rightmost column. That is, it incorrectly says that ***is*** and ***are*** are equally likely to be the next word. This result shows that the model's ability to get the right answer for this row depends on having access to U1 or U2.

N5. On your Answer Sheet, provide values for the empty cells in the table. Notes:

- For space (e), your answer should be 5 words long; there are many possible correct answers, but you only need to provide one.
- In the second-to-last row, the correct answer is 50%/50% because either option is a possible way to continue the sentence. ***is*** could be used for a phrase like ***the dog by the zebras that is wagging its tail*** (where ***that is wagging its tail*** describes ***the dog***), and ***are*** could be used for a phrase like ***the dog by the zebras that are galloping*** (where ***that are galloping*** describes ***the zebras***).

Start of sentence	**Correct answer**	**Computer predictions with...**			
		U1 on, U2 on	**U1 off, U2 on**	**U1 on, U2 off**	**U1 off, U2 off**
the dogs	is: 0% are: 100%	is: 0% are: 100%	is: 0% are: 100%	is: 0% are: 100%	is: 50% are: 50%
the dog	is: 100% are: 0%	is: 100% are: 0%	is: 100% are: 0%	is: 100% are: 0%	is: 50% are: 50%
the dog by the zebras	is: 100% are: 0%	is: 100% are: 0%	is: 0% are: 100%	is: 100% are: 0%	is: 50% are: 50%
the dog that the zebras	is: 0% are: 100%	is: 0% are: 100%	is: 0% are: 100%	is: 50% are: 50%	is: 50% are: 50%
the dog that	is: 100% are: 0%	is: 100% are: 0%	is: 100% are: 0%	is: 50% are: 50%	is: 50% are: 50%
the dog by the zebras that	is: 50% are: 50%	**(a)**	**(b)**	**(c)**	**(d)**
(e)	is: 0% are: 100%	is: 100% are: 0%	is: 100% are: 0%	is: 100% are: 0%	is: 50% are: 50%

<u>Make sure you record your answers in your Answer Sheets!</u>